Workers Compensation Claims Prediction

A data-driven approach to identify long-duration claims early

SUBMITTED BY
Mohd Kaunain

MENTOR
Bhargav Dakka

ORGANIZATION
Sure Trust

Agenda

Client Context: RWST

Return to Work & Support Team

RWST manages claims extending beyond **12 weeks**. During incapacity, they cover wages, medical services, and rehabilitation.

Current State: Claims extending beyond **39 weeks** generate significant losses. RWST currently intervenes reactively, often too late.

Problem Statement

The 39-Week Threshold

Claims exceeding 39 weeks are the critical tipping point for financial loss.

Predictive Goal

Identify high-risk claims (>39 weeks) early in their lifecycle.

Business Impact

Reduce massive payouts at 130 weeks by prioritizing support strategically.

Project Objectives

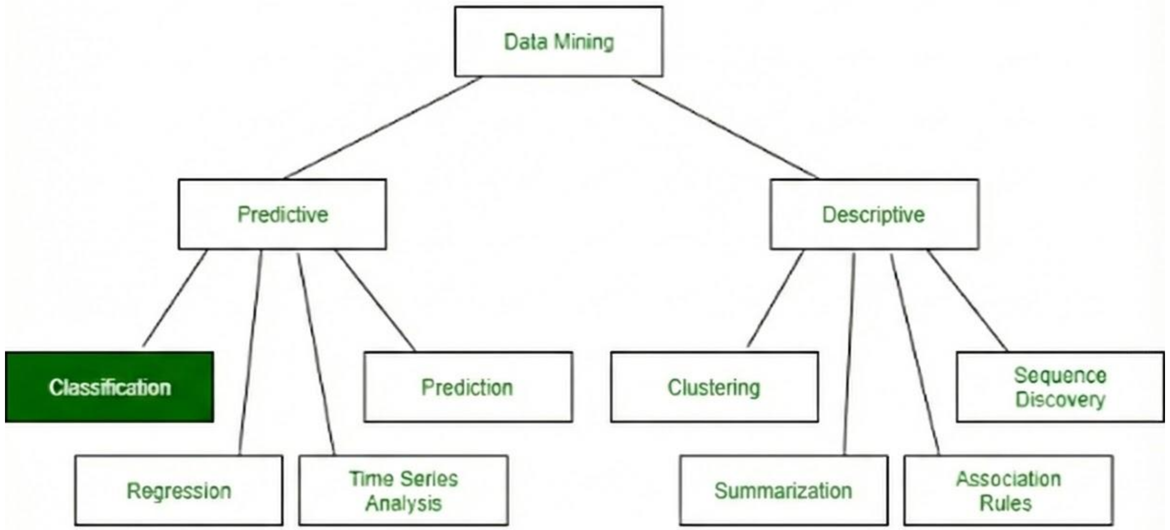
- **Classification Modeling:** Predict if claim duration will exceed 39 weeks.
- **Pattern Analysis:** Analyze financial payment patterns in early claim stages.
- **Driver Identification:** Identify top drivers (medical, rehab, etc.) of long-duration claims.
- **Actionable Insights:** Provide RWST with data for proactive decision-making.

Dataset Overview

Source: Workers_Compensation_Claims_data.csv

Volume: 28,500 Rows | 22 Columns

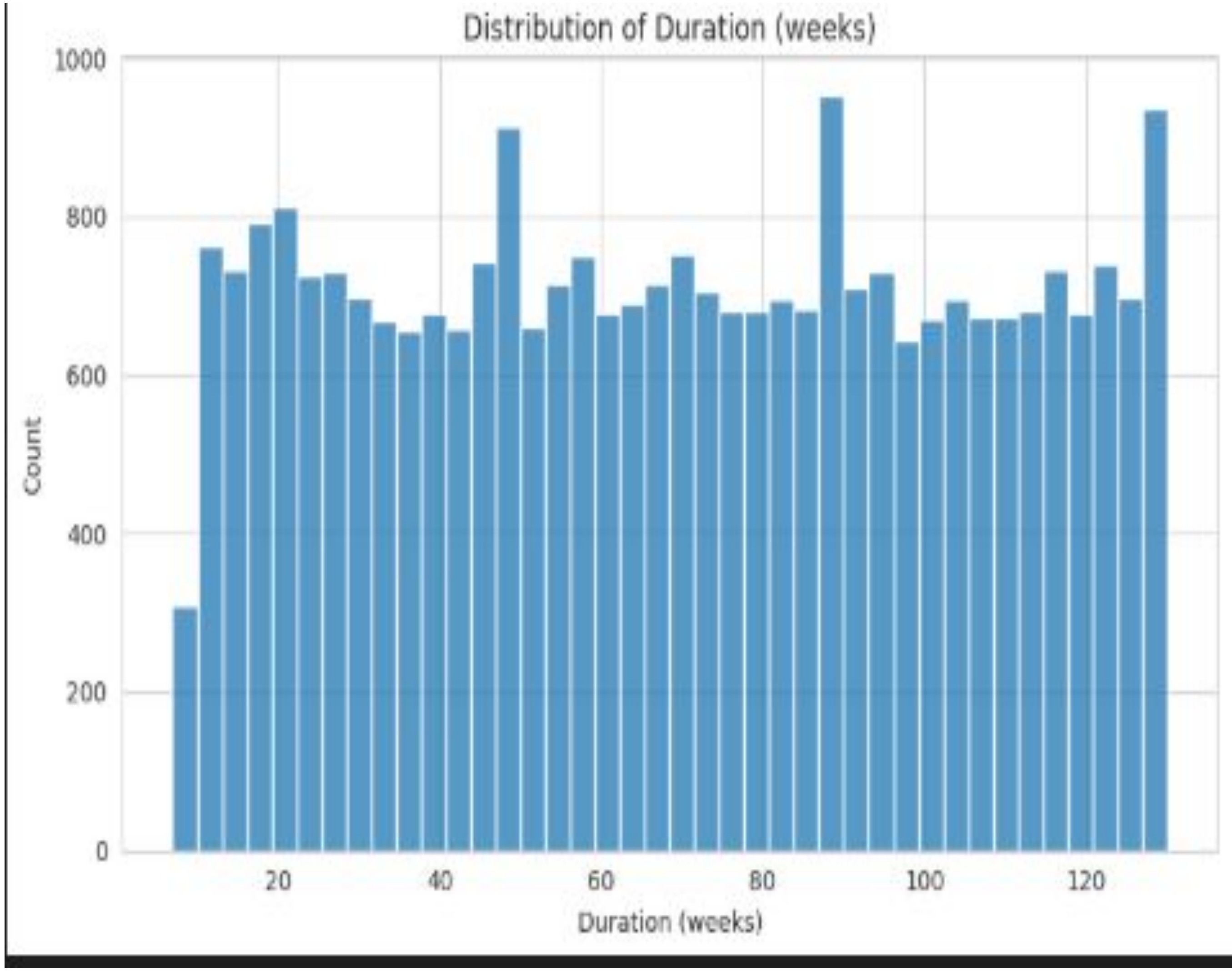
- **Claim Details:** Claim Number, Type, Status
- **Demographics:** Age, WPI (Impairment Score)
- **Financials:** Gross Paid, Rehab, Medical, Net Paid
- **Target:** Duration (weeks)



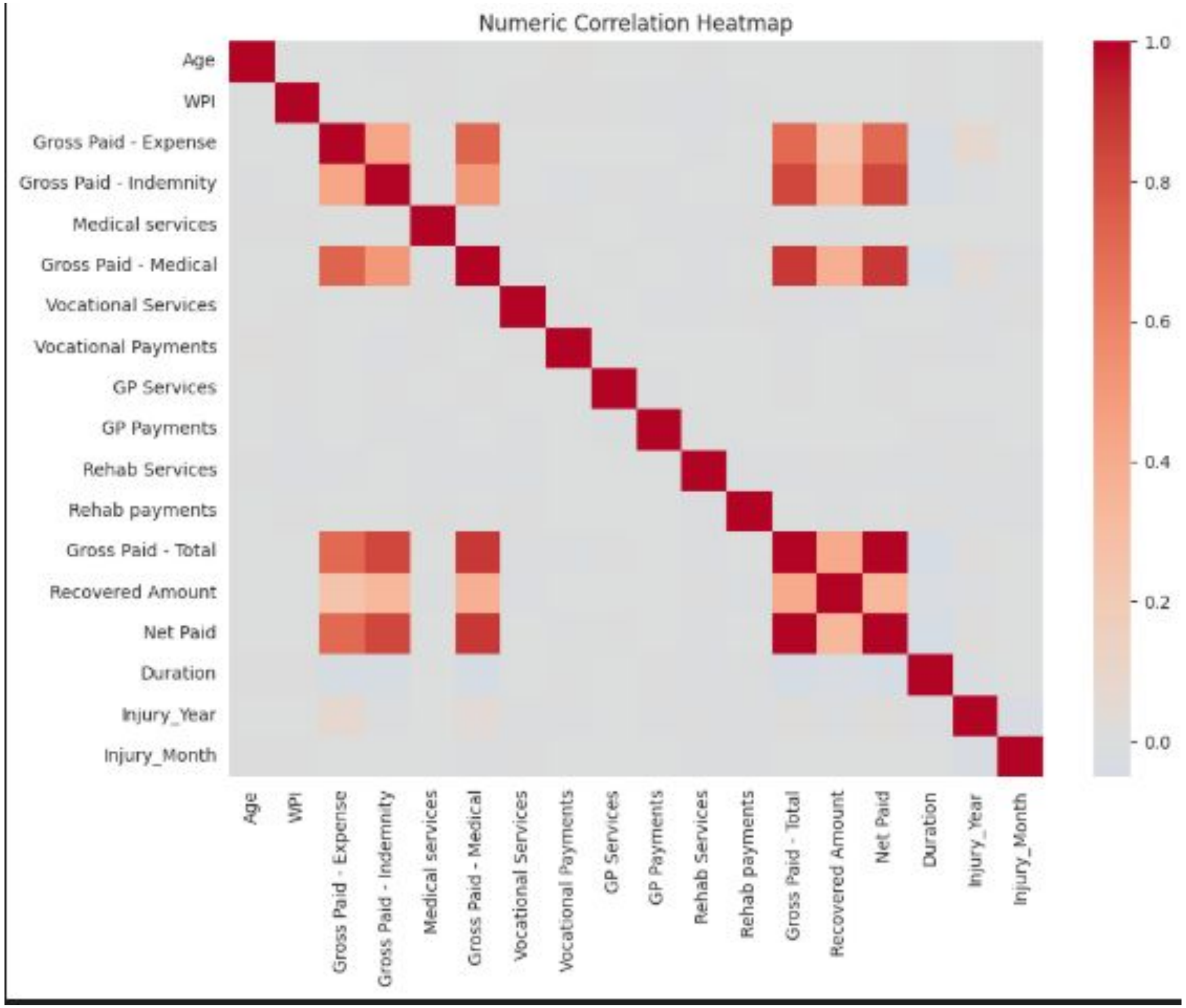
EDA: Duration Distribution

Insight: The distribution is balanced. Sampling techniques equalized classes, ensuring the model learns long-duration claims (>39 weeks) as effectively as short claims.

- Most claims settle within **0–40 weeks**.
- The "Long Duration" (>39 weeks) claims are the minority class but represent the highest cost risk.
- This imbalance necessitates class-balancing techniques in modeling.



EDA: Correlation Analysis



Observations:

- **Weak Single Predictors:** Simple analysis shows no single financial feature has a strong *linear* relationship with duration.
- **WPI Impact:** Initial impairment scores (WPI) show a surprisingly low direct correlation with total time off, suggesting other factors are more influential.
- **Multivariate Need:** This indicates that complex, non-linear relationships and feature interactions likely drive long-duration claims, which a simple analysis cannot capture.

Modeling Strategy

✕Regression (Discarded)

Goal: Predict exact weeks.

Outcome: Failed (Negative R^2). Duration is too volatile to predict as a continuous variable with high precision.

✓Classification (Selected)

Goal: Predict High Risk (>39 weeks).

Outcome: Successful. We used **RandomForestClassifier** with class weights to prioritize recall (catching risky claims).

Model Performance

99%
Recall
Captures almost all high-risk claims.

74%
Accuracy
Overall correctness.

0.85
F1 Score
Balanced performance.

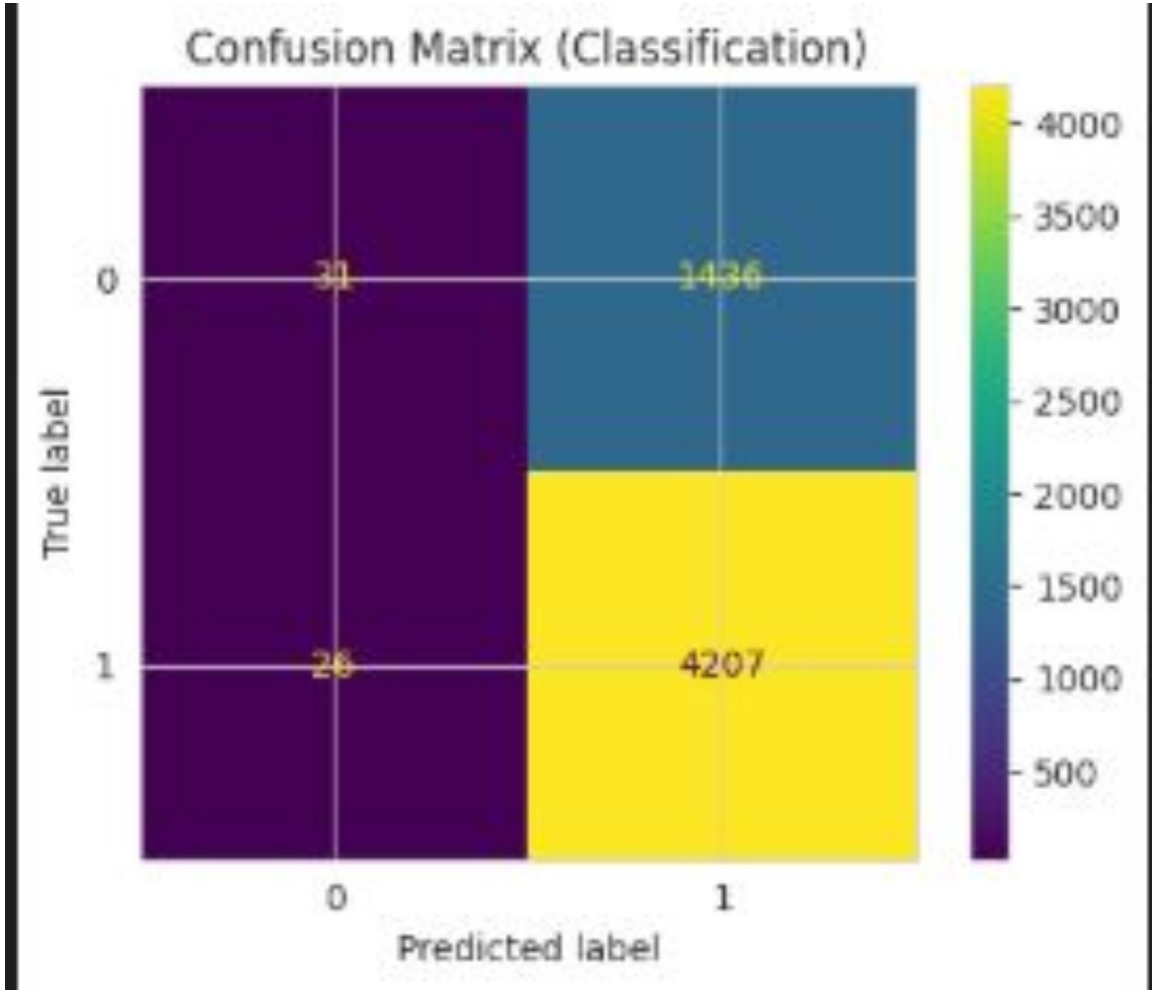
Confusion Matrix

Visual Analysis:

True Positives (High Risk Detected): The model correctly identifies the vast majority of claims that actually go over 39 weeks.

False Negatives (Missed Risk): Extremely low. This is crucial—we rarely miss a dangerous claim.

Strategic Trade-off: The model generates **1,436 False Positives** (predicting risk where there is none). This is intentional: we prioritized a hypersensitive model to ensure 99% of actual high-risk cases are caught, even if it means flagging some safe claims

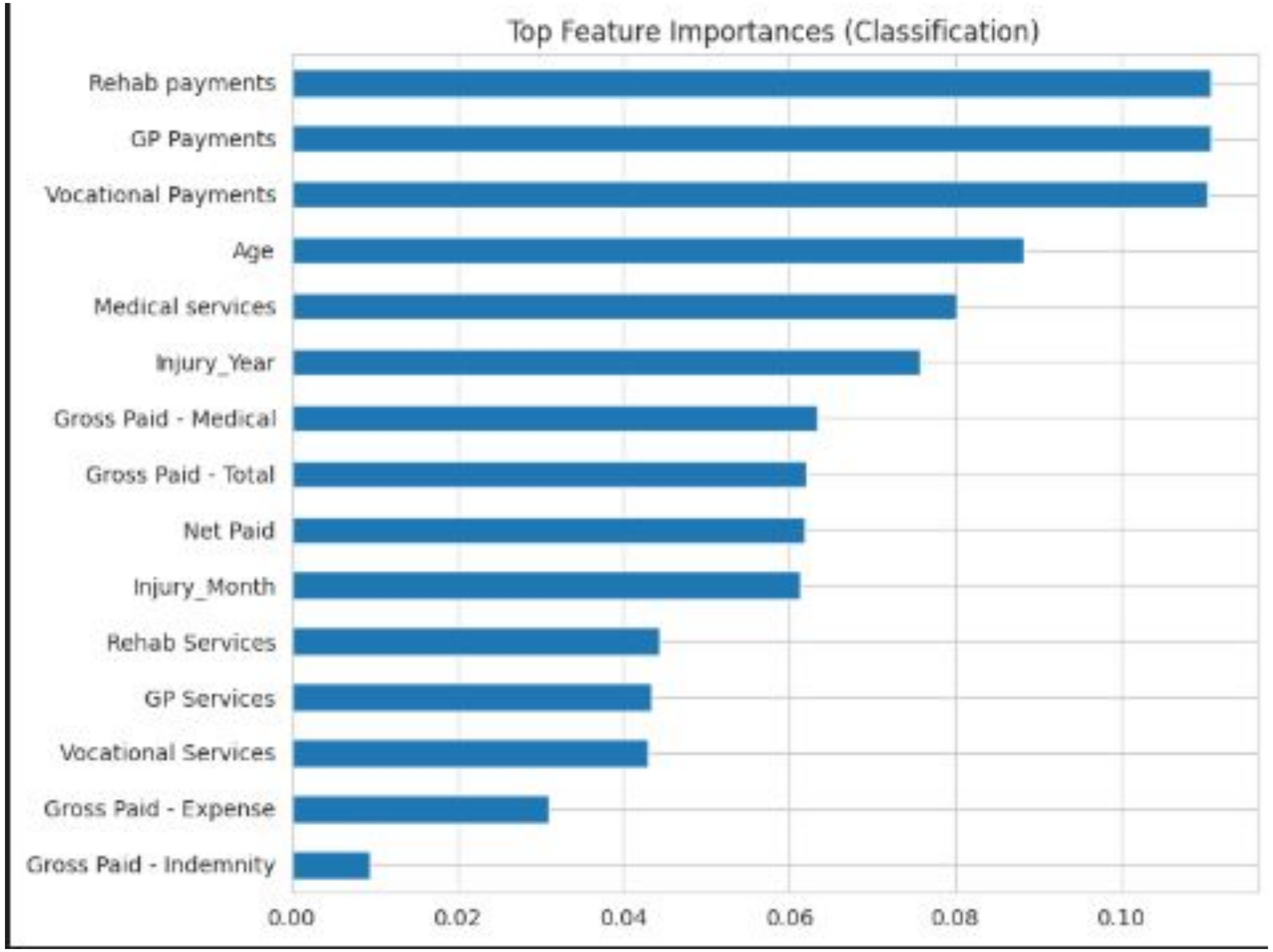


Feature Importance Drivers

Top Drivers of Long-Duration Claims

The most critical predictor identified by our model is the type of early spend:

- **Rehab & GP Payments:** Significant early investment in rehabilitation and General Practitioner services strongly signals a complex, long-duration claim.
- **Vocational Payments:** Costs related to return-to-work support are a key indicator of extended time off.
- **Medical Services:** A high volume of medical services is another strong driver of long duration.



Recommendations for RWST

Target Early Spend

Investigate claims with abnormally high early Rehab or GP payments immediately.

Dashboarding

Deploy a Power BI dashboard to auto-flag claims predicted as "High Risk" (>39 weeks).

Proactive Support

Shift resources from reactive administration to proactive worker support for flagged cases.

Thank You